# 1 Introduction

## 1.1 Definition of masting

1. Observations of masting in relation to seed predation and flowering

## 1.2 Major hypotheses for masting

1. Predation satiation
2. Pollination coupling
3. Resource matching

## 1.3 Hypotheses operating at different reproductive stages like filters

1. Pollination success (Pollination coupling)
2. Resources availability (Resource Matching)
3. Dispersal (Predation satiation)
4. Mechanisms are not mutually exclusive
5. Traits thus can be important to understand masting through their relative importance

## 1.4 Trait-based perspective

1. Literature review on understanding masting from traits perspective
    - Larger seed is associate with higher CV
    - Animal dispersed seeds have higher CV than wind dispersed seeds
    - Wind pollination is related to higher CV
    - Recent papers also focus on resource perspective using leaf and shoot traits
2. Existing studies seem focus mainly on correlations
3. Not clear connection between traits and hypotheses and potential evolutionary drivers

### 1.5 Introduce framework linking hypothesis x traits x masting

1. Link traits to hypotheses help understand drivers of masting for different species
2. If the pattern is strong enough, we might be able to find it even using a binary masting definition
3. Here we use North American tree data to test this framework since they have information on their life-history
    - Masting info available
    - Includes both masting and non-masting species
    - Not using CV data because don't have stand-level traits data

## 2 Methods

### 2.1 Data collection

1. Data scraping
    - Trait selection
    - Data sources: Silvics of North America, Kew Seed Dataset, TRY Dataset
    - Keywords used for data scraping
        - Definition of strong masting vs. non-masting (referred to as masting vs. non-masting moving forward)
        - Criteria for other traits
2. Data cleaning
    - Cleaning of categorical traits
    - Decision for numeric traits with a range

### 2.2 Statistical analysis

1. Main analysis for masting pattern and single trait: phyloglm
2. Masting pattern with a multivariate analysis: NMDS
3. Phylogenetic signal check for traits: Pagel's $\lambda$ and D-statistics

# 3 Discussion

## 3.1 Summary of main findings

1. Seed weight supports the predation satiation hypothesis
2. Pollination supports the pollination coupling hypothesis
3. These act at the start and the end of the reproductive cycle, provide evolutionary incentive
4. Discuss mechanisms in the context of existing literature
   - Previous literature on seed mass and masting think from the resource depletion perspective
   - From evolutionary perspective, seeds attractive to animal dispersers will benefit more from masting
5. Other traits show some pattern but statistically non-significant
6. Transition to alternative hypotheses

## 3.2 Masting is a population-level phenomenon

1. No evidence found at species level data does not mean traits are not related to masting
2. Examples of species that mast only under certain environmental conditions (E.g., *Hymenaea coubaril*)
3. Traits need to be measured locally

## 3.3 Different species selected under different hypotheses

1. Introduce flowering masting vs. seed masting
2. Different species driven by different mechanisms
3. Need a more refined definition of masting and additional flowering-related traits
4. Transition to phylogeny

## 3.4 Evolutionary scale

1. Angiosperm vs. gymnosperm results
   - More masting gymnosperm sp
   - Longer leaf longevity in gymnosperm
   - More monoecious species in gymnosperm

- Wind pollination is universal among gymnosperms
- Gymnosperm and angiosperm have distinct trait space occupancy
- Importance of looking for potential evolution support between gymnosperm and angiosperm
  - Test if masting was gradually diluted in angiosperm as they transitioned to biotic pollination and faster life histories

2. Many traits have strong phylogenetic signal, it is hard to tease out traits from masting, which limits the inference
3. Data is strongly nested in phylogeny, and data is sparse.
4. Importance of testing within clades and collect more data (*Quercus.*)